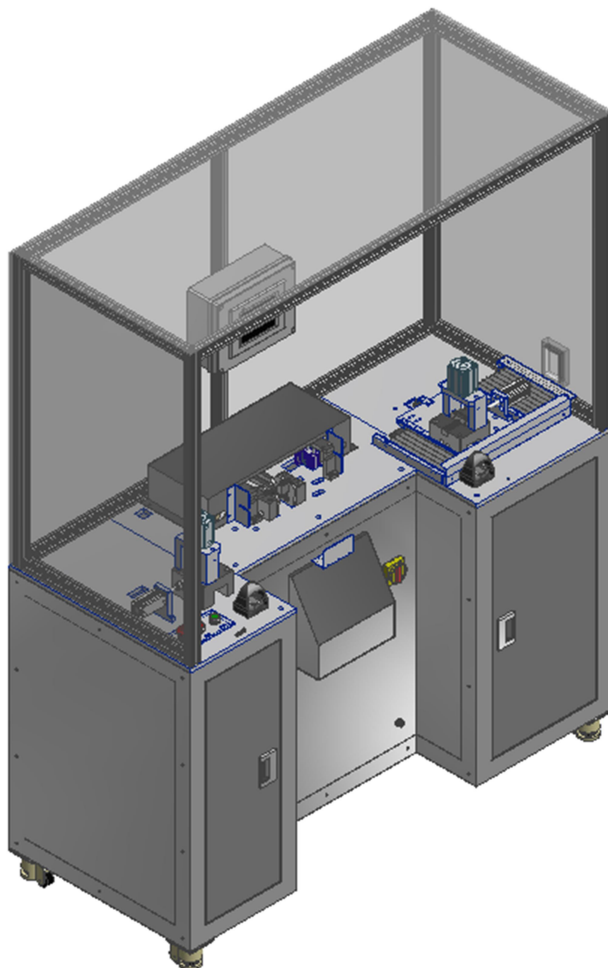




CRIMPING MACHINE

USER MANUAL



OPERATION MANUAL

MACHINE OPERATION PROCEDURE

1. TURN THE POWER SWITCH .
2. WAIT UNTIL TOUCH PANEL DISPLAY IS ON AND MAKE SURE EMERGENCY BUTTON IS IN OFF POSITION.
3. PRESS RESET BUTTON AT TOUCH PANEL DISPLAY AND MACHINE WILL PERFORM HOMING ROUTINE.
4. SET MACHINE TO AUTO MODE.
5. FOR STEP 1, PUT CLIPS TO BOTH HOLDERS AND THEN THE BELT.
6. HOLD THE BELT BY BOTH HANDS WHILE MAKING SURE THAT THE HANDS IS NEAR THE SAFETY SENSOR.
7. PRESS THE FOOT SWITCH, TO FINISH STEP 1.
8. FOR STEP 2, PUT THE BELT TO LEFT AND RIGHT HOLDERS.
9. START STEP 2 BY PUTTING FINGERS AT BOTH TOUCH SWITCH.
10. AFTER FINISH STEP 2, MEASURE THE BELT LENGTH AT THE MARKER.

MACHINE FRONT PANEL

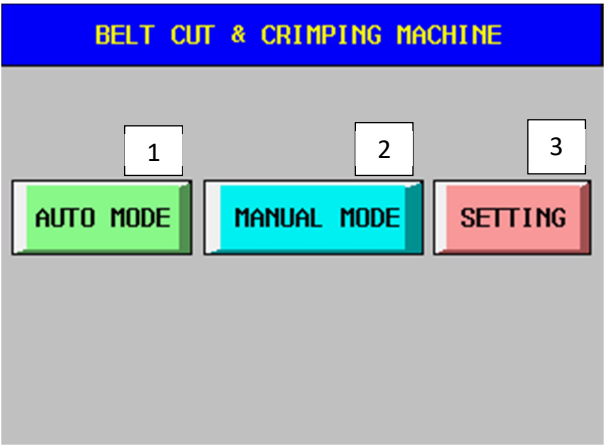


Figure 1: Main panel

MAIN PANEL

No	Function	Description
1	AUTO MODE	When turned on, the machine is in auto mode and ready for production. When turned off, the machine is in debugging or repairing mode.
2	MANUAL	TO enter IO & actuator setting mode.
3	SETTING	To enter setting mode.

AUTO FUNCTION

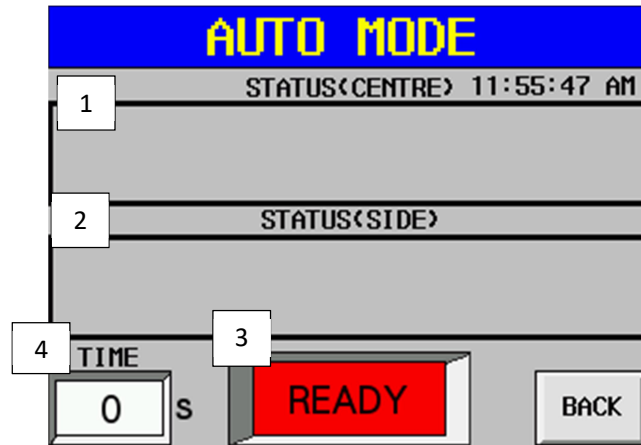


Figure 2: Auto Screen

AUTO SCREEN

No	Function	Description
1	STATUS 1	Machine status for cutting and crimping.
2	STATUS 2	Machine status for crimping and stretching.
3	READY	Machine ready indicator and button.
4	TIMER	Cycle time for full process

SETTING FUNCTION

The screenshot shows a screen titled "SETTING 1" in yellow text on a blue background. Below the title, there are five rows of settings, each with a label, a numeric input field, and a unit "ms". The settings are: CRIMP CENTRE DELAY, CRIMP SIDE DELAY, CUT DELAY, TOP PRESS DELAY, and STRECTH TIMEOUT. Each input field contains the number "0". A "BACK" button is located at the bottom right of the screen.

Setting Label	Value	Unit
CRIMP CENTRE DELAY	0	ms
CRIMP SIDE DELAY	0	ms
CUT DELAY	0	ms
TOP PRESS DELAY	0	ms
STRECTH TIMEOUT	0	ms

BACK

Figure 3: Setting Screen

SETTING SCREEN

No	Buttons	Description
1	CRIMP CENTRE DELAY	Crimp centre delay duration setting.
2	CRIMP SIDE DELAY	Crimp side delay duration setting.
3	CUT DELAY	Cut delay duration setting.
4	TOP PRESS DELAY	Top press delay duration setting.
5	STRECTH TIMEOUT	Belt stretch timeout duration setting.

MANUAL FUNCTION

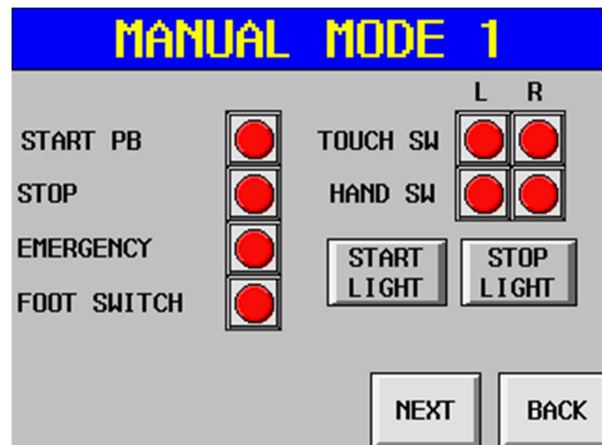


Figure 4: Input Screen

INPUT SCREEN

No	Function	Description
1	INPUT	To check the sensor/input signal.

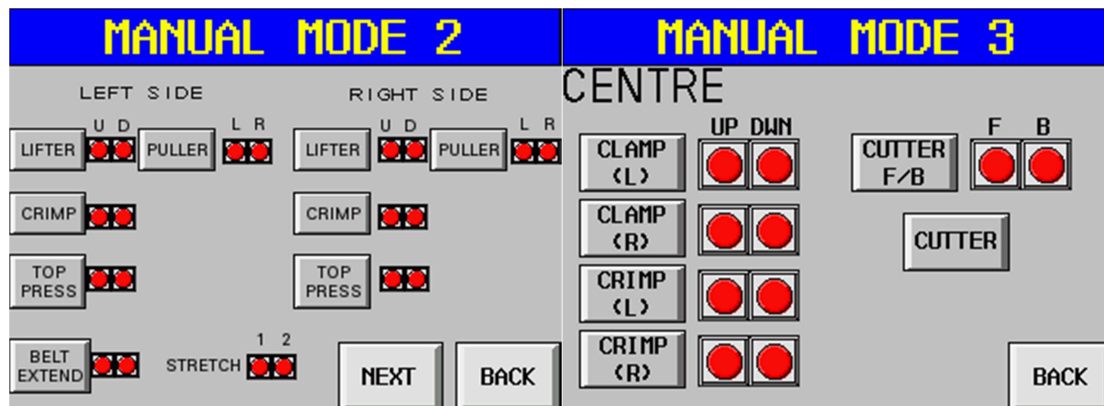
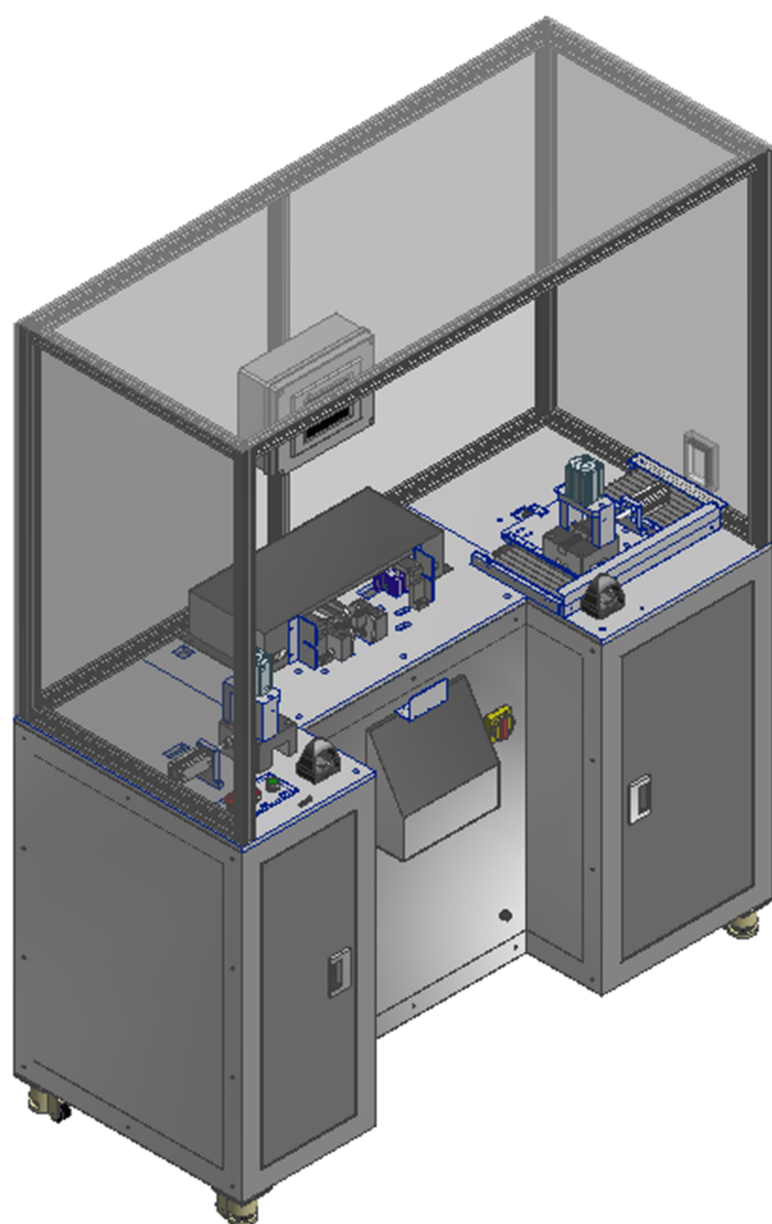


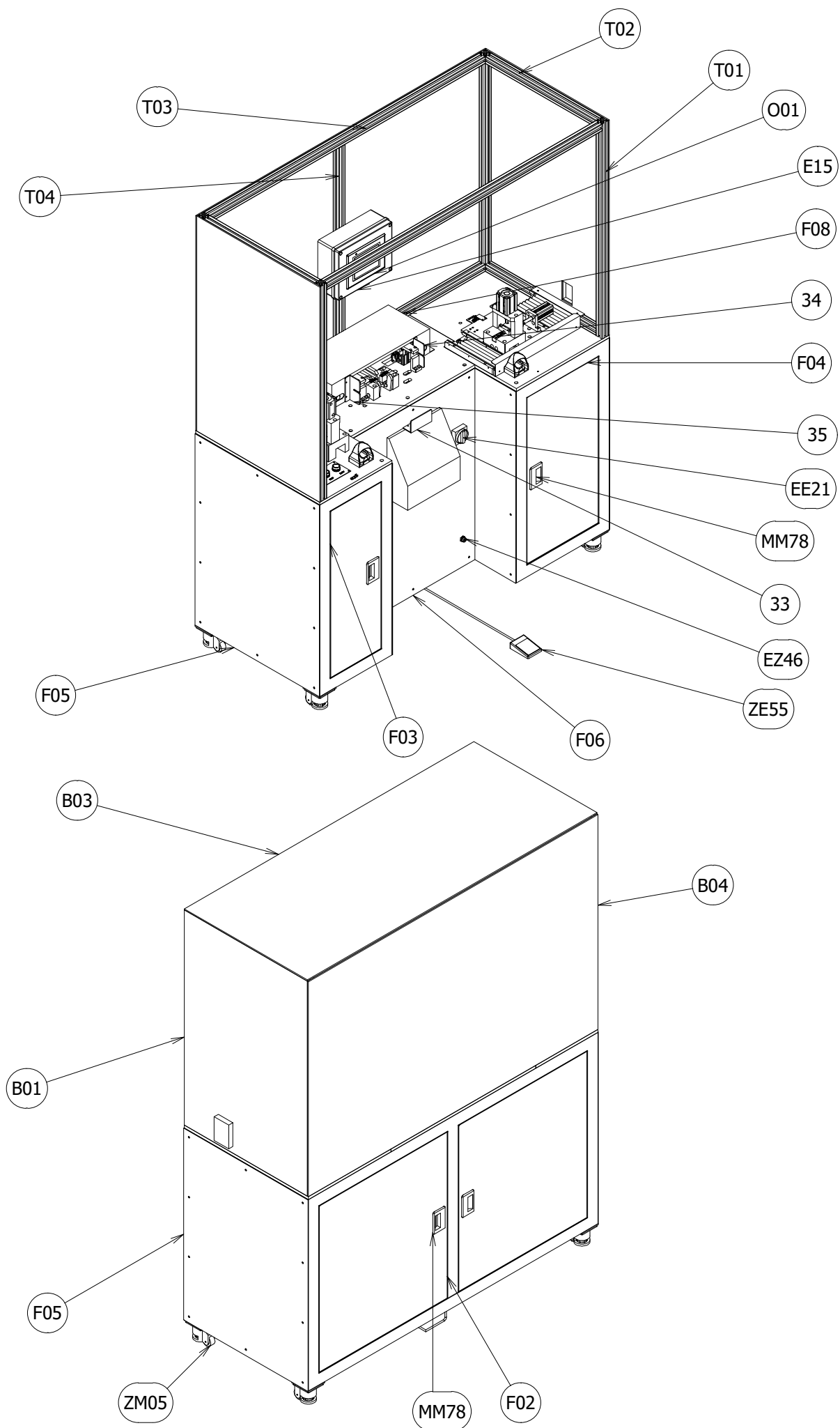
Figure 5: Output Screen

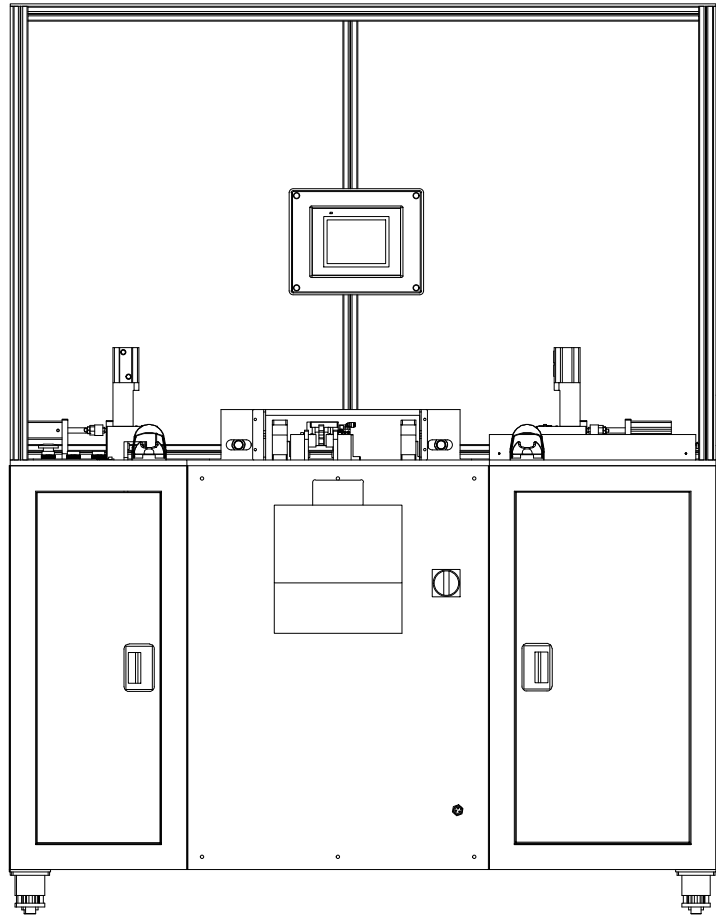
OUTPUT SCREEN

No	Function	Description
1	OUTPUT	Manual trigger output.

MECHANICAL PART LIST







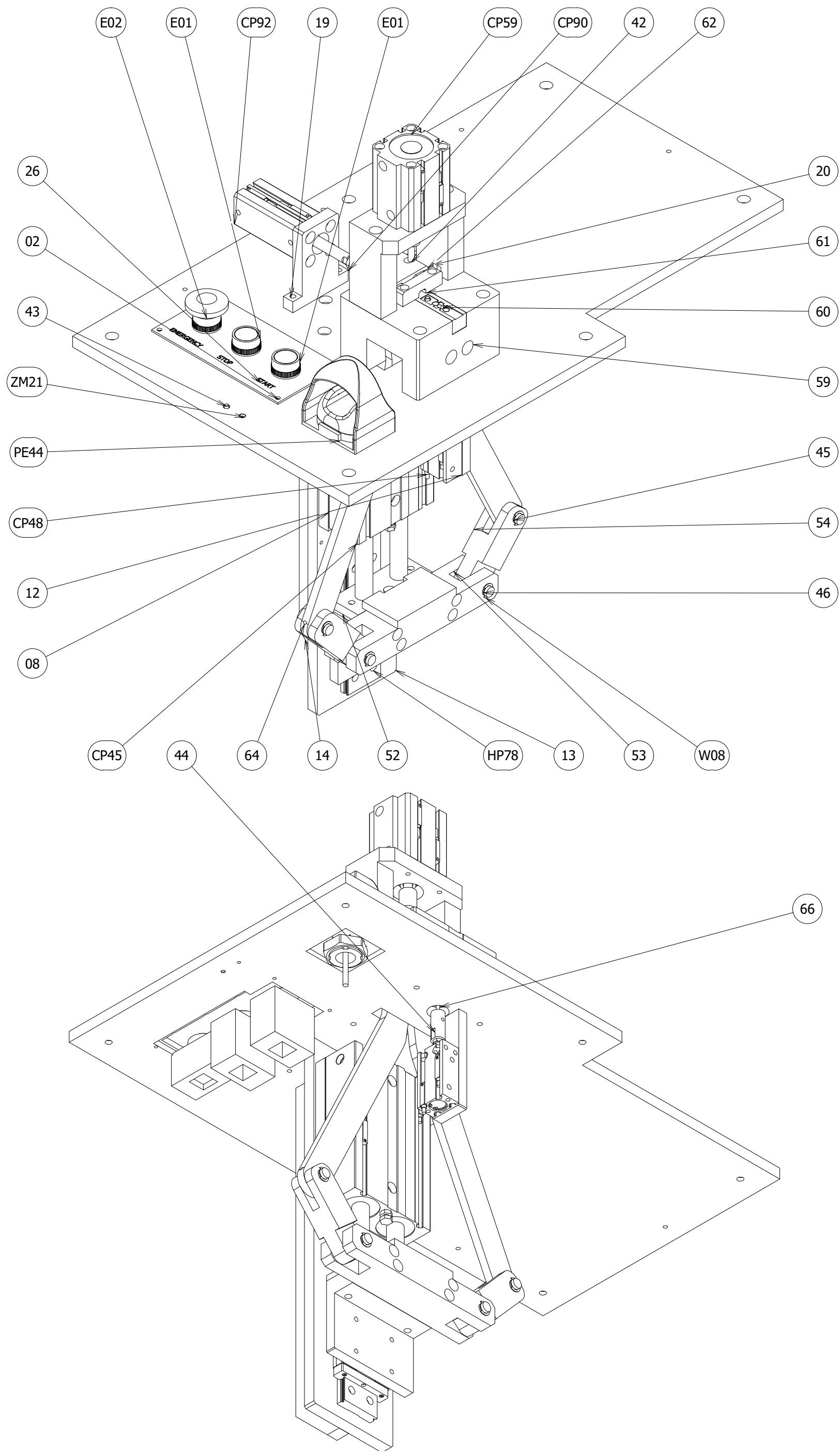
PARTS LIST	
ITEM No	PART NUMBER
33	BELT L 1
34	BELT L 2
35	BELT L 3

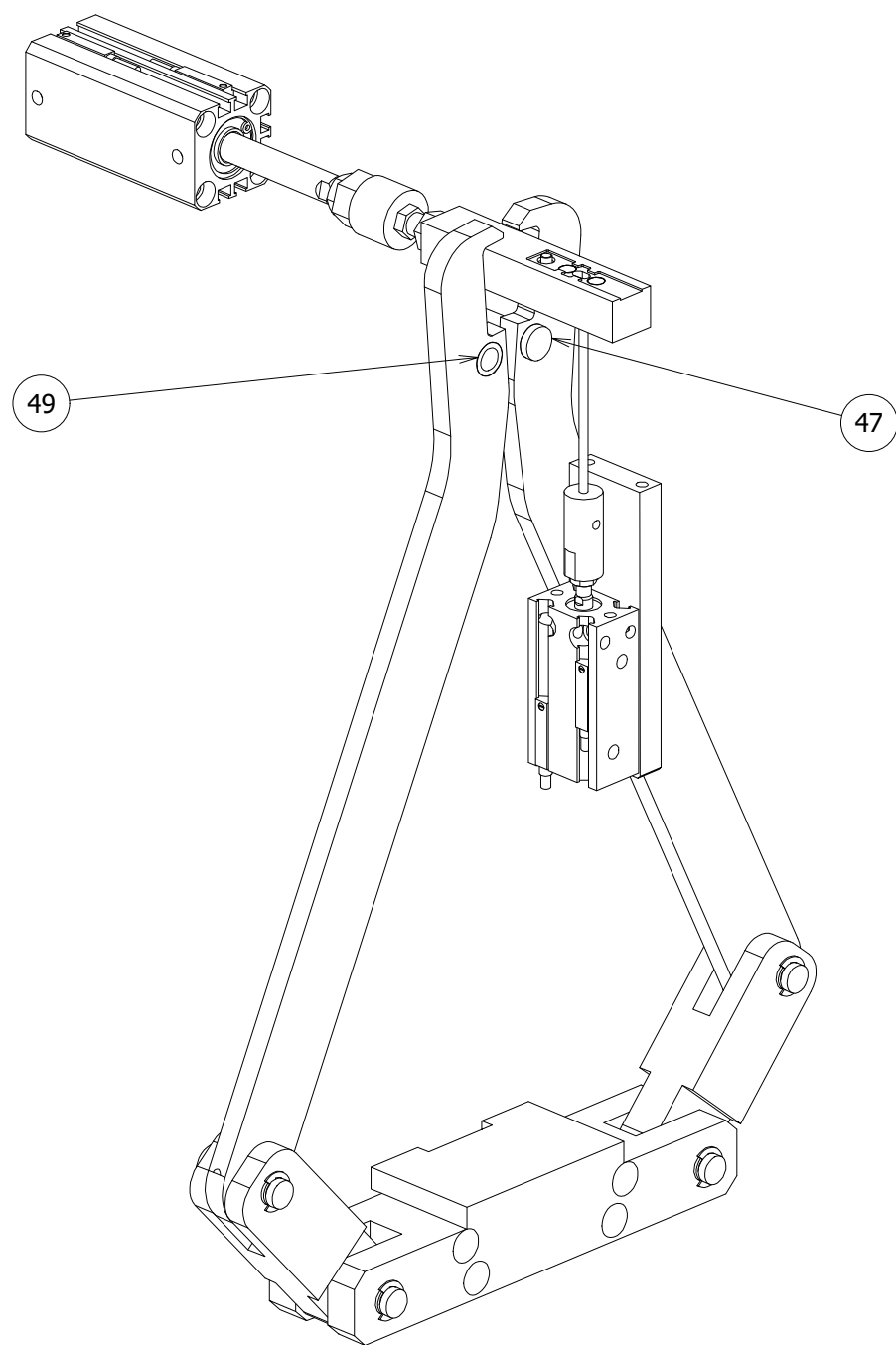
PARTS LIST	
ITEM No	PART NUMBER
F01	F PLATE
F02	REAR DOOR
F03	FRONT DOOR L
F04	FRONT DOOR R
F05	SIDE COVER 1
F06	FRONT COVER
F07	BOTTOM PLATE
F08	TOP COVER

PARTS LIST	
ITEM No	PART NUMBER
T01	AL 3030 1
T02	AL 3030 2
T03	AL 3030 3
T04	AL 3030 4
T05	AL 3030 2B
T06	AL 3030 3B

PARTS LIST	
ITEM No	PART NUMBER
B01	SIDE COVER R
B02	SIDE COVER L
B03	T COVER
B04	B COVER
B05	SIDE COVER R2

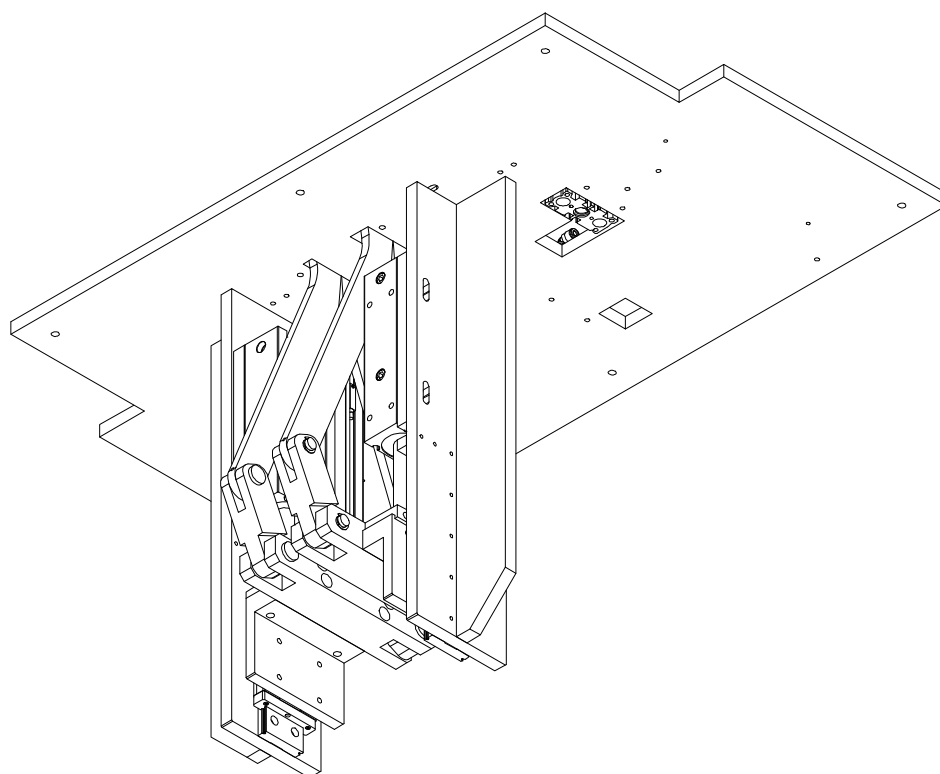
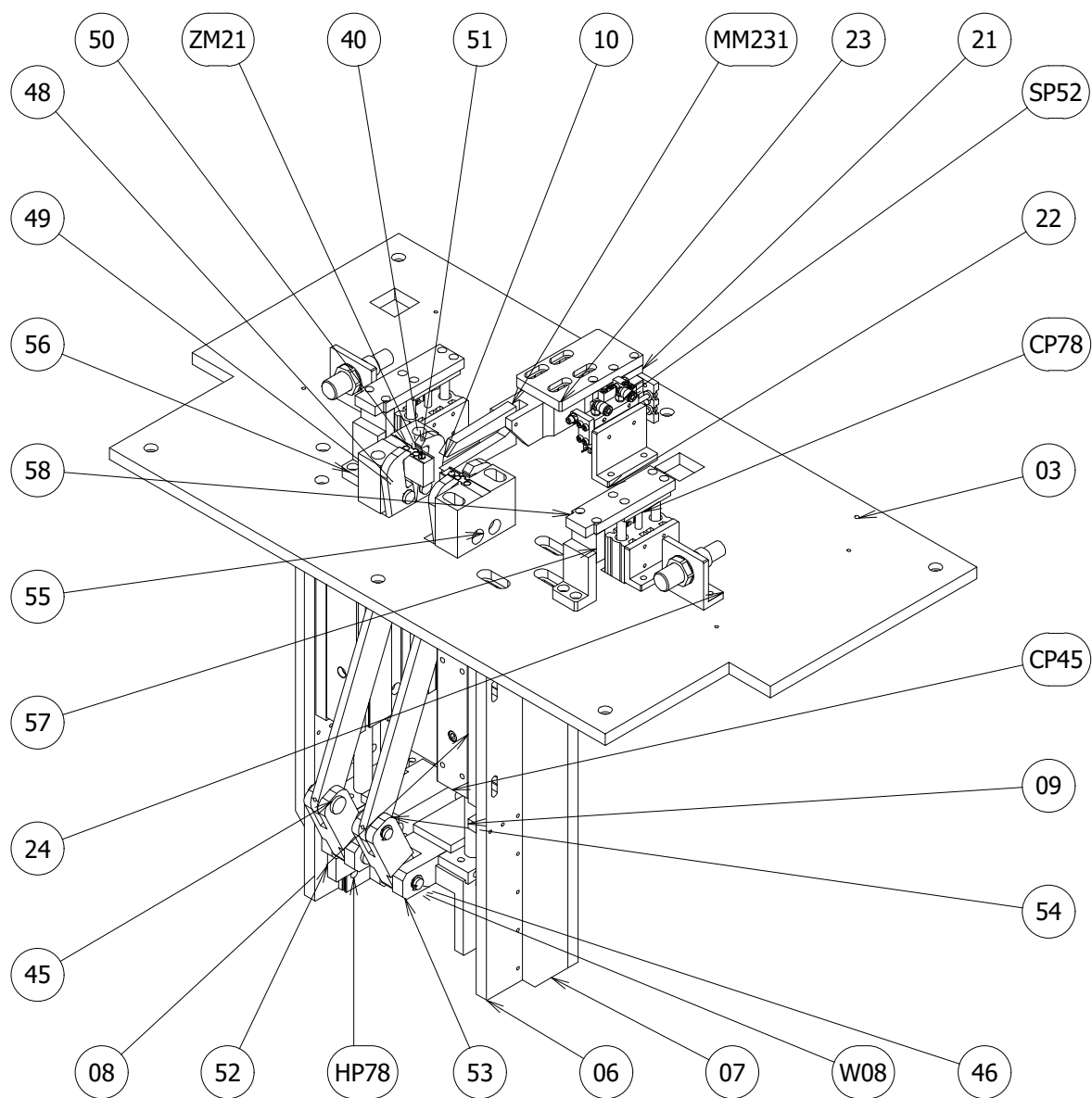
PARTS LIST	
ITEM No	DESCRIPTION
E15	PLASTIC BOX
EE21	CAM SW ON OFF
EZ46	EST THREE PIN MALE
F01	CASTER PLATE
O01	TOUCH PANEL DISPLAY
PE44	TOUCH SWITCH
ZE55	FOOT SWITCH
ZM05	CASTER WITH LOCK





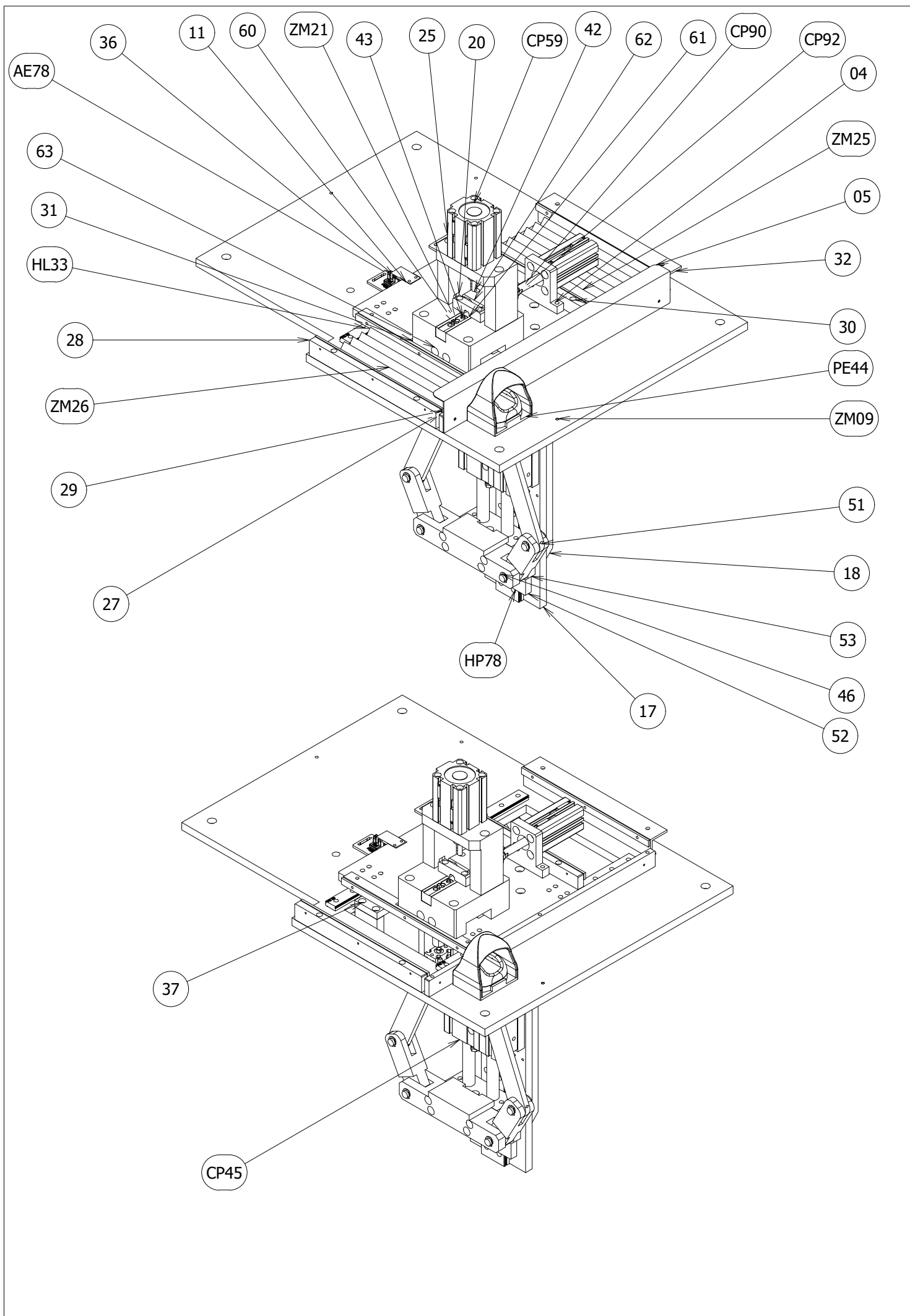
PARTS LIST	
ITEM No	PART NUMBER
02	PLATE L
08	CYL PLATE 1
12	SMG PLATE 1
13	STAND 1L
14	STAND 2L
19	SSD L1
20	SSD BLOCK 1
26	S PLATE
42	PUSHER PIN
43	G PIN2
44	PUSHER PIN 2 L
45	L PIN 1
46	L PIN 2
47	L PIN 3
49	COLLAR 1
52	M L1
53	M L2
54	C2
59	M BASE L
60	P SLIDER 1
61	B PUSHER 1
62	B PUSHER 2
64	C1 A1
66	EJECTOR SHAFT D3

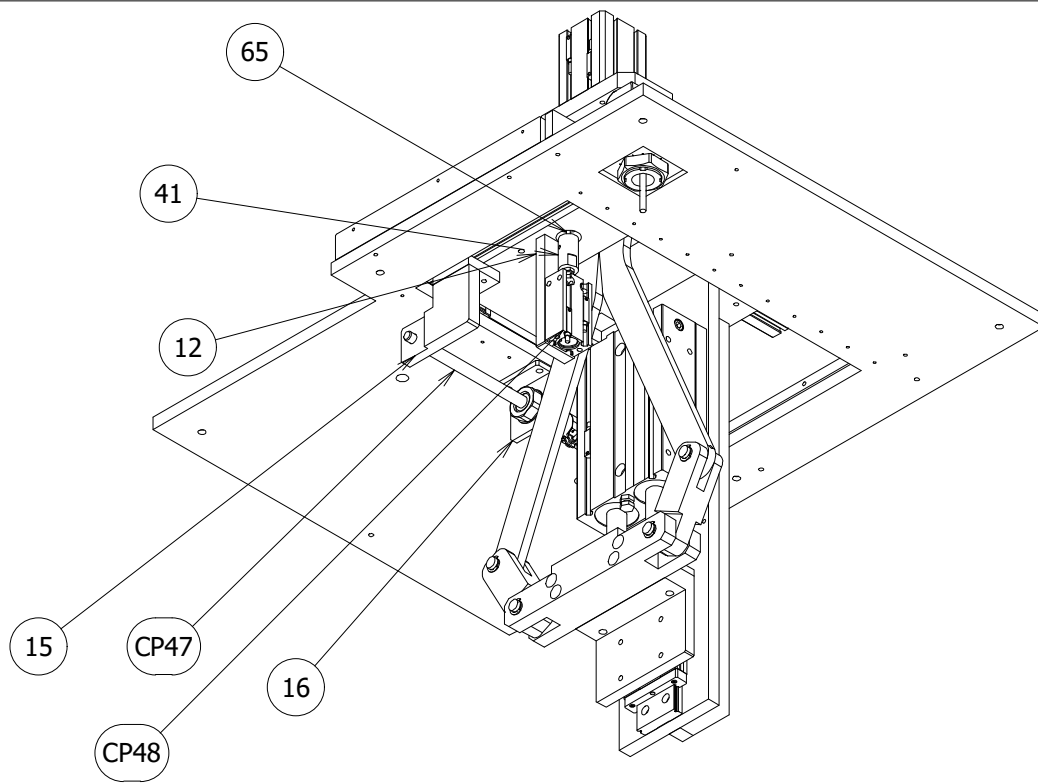
PARTS LIST	
ITEM No	DESCRIPTION
CP45	DUAL ROD PNEUMATIC CYLINDER
CP48	GUIDED COMPACT CYLINDER
CP59	PNEUMATIC COMPACT CYLINDER
CP90	FLOATING JOINT
CP92	PNEUMATIC COMPACT CYLINDER
E01	ON SWITCH
E01	ON SWITCH
E02	ON SWITCH
HP78	LINEAR SLIDE TABLE
PE44	TOUCH SWITCH
W08	E CLIP
ZM21	MAGNET



PARTS LIST	
ITEM No	PART NUMBER
03	PLATE M
06	STAND 1
07	STAND 2
08	CYL PLATE 1
09	STOPPER
10	CYL PLATE 2
21	AC L1
22	CYL PLATE 3
23	AC L2
24	CYL PLATE 4
40	G PIN 1
45	L PIN 1
46	L PIN 2
48	L PIN 4
49	COLLAR 1
50	M BASE
51	C1
52	M L1
53	M L2
54	C2
55	M BASE 2
56	B HOLDER
57	B HOLDER 2
58	B CLAMP

PARTS LIST	
ITEM No	DESCRIPTION
CP45	DUAL ROD PNEUMATIC CYLINDER
CP78	GUIDED COMPACT CYLINDER
HP78	LINEAR SLIDE TABLE
MM231	AIR SCISSORS
SP52	PNEUMATIC SLIDE TABLE
W08	E CLIP
ZM21	MAGNET





PARTS LIST	
ITEM No	PART NUMBER
01	PLATE R
04	SSD R1
05	R COVER L
08	CYL PLATE 1
11	SLIDE PLATE R
12	SMG PLATE 1
15	CMK2 L2
16	CMK2 L1
17	STAND 1r
18	STAND 2r
20	SSD BLOCK 1
25	SEN L2
27	STOPPER 2
28	STOPPER 3
29	STOPPER 4
30	STOPPER 5
31	STOPPER 6
32	STOPPER 7
36	SEN L1
37	ADD PLATE 2
41	PUSHER PIN 2 R
42	PUSHER PIN
43	G PIN2
45	L PIN 1
46	L PIN 2
47	L PIN 3
49	COLLAR 1
51	C1
52	M L1
53	M L2
54	C2

PARTS LIST	
ITEM No.	PART NUMBER
60	P SLIDER 1
61	B PUSHER 1
62	B PUSHER 2
63	M BASE R
65	EJECTOR SHAFT R1

PARTS LIST	
ITEM No	DESCRIPTION
AE78	PHOTOELECTRIC SENSOR
CP45	DUAL ROD PNEUMATIC CYLINDER
CP47	PNEUMATIC ROD CYLINDER
CP48	GUIDED COMPACT CYLINDER
CP59	PNEUMATIC COMPACT CYLINDER
CP90	FLOATING JOINT
CP92	PNEUMATIC COMPACT CYLINDER
HL33	LINEAR SLIDE TABLE
HP78	LINEAR SLIDE TABLE
PE44	TOUCH SWITCH
W08	E CLIP
ZM09	MAGNET
ZM21	MAGNET
ZM25	RUBBER COVER
ZM26	RUBBER COVER

PLC I/O AND ELECTRICAL
WIRING DIAGRAM

CRIMPING MACHINE PLC I/O

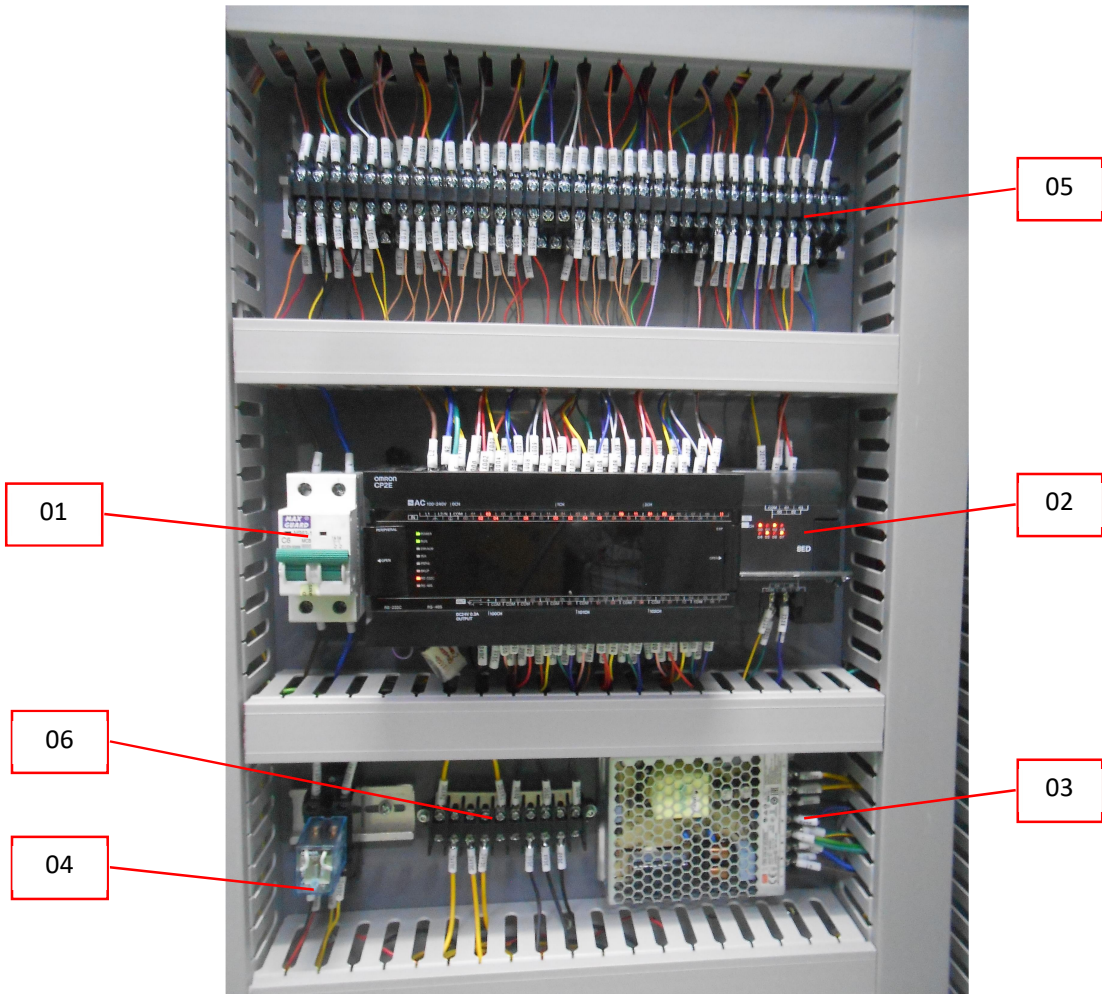
Model	PLC Mark	Label	Description	Remark	Device Pin
CP2E-S60DR-A	L1	L1	AC Live		AC Live
	L2/N	N1	AC Neutral		AC Neutral
	E	E1	AC Earth		AC Earth
	COM	24Vdc	24V Power supply		24Vdc
0CH (INPUT)	00	X000	START		
	01	X001	STOP		
	02	X002	EMERGENCY		
	03	X003	TOUCH SW LEFT		
	04	X004	TOUCH SW RIGHT		
	05	X005	FOOT SW		
	06	X006	HAND SW		
	07	X007	HAND SW		
	08	X008	STRETCH1 } U-SENSOR		
	09	X009	STRETCH2 }		
	10	X010			
	11	X011			
1CH (INPUT)	00	X100	LEFT } PULLER LEFT		CYLINDER
	01	X101	RIGHT }		SENSOR
	02	X102	LEFT } PULLER RIGHT		CYLINDER
	03	X103	RIGHT }		SENSOR
	04	X104	UP } TOP PRESS L		CYLINDER
	05	X105	DOWN }		SENSOR
	06	X106	UP } TOP PRESS R		CYLINDER
	07	X107	DOWN }		SENSOR
	08	X108	UP } CRIMP LEFT		CYLINDER
	09	X109	DOWN }		SENSOR
	10	X110	UP } CRIMP RIGHT		CYLINDER
	11	X111	DOWN }		SENSOR
2CH (INPUT)	00	X200	UP } LIFTER LEFT		CYLINDER
	01	X201	DOWN }		SENSOR
	02	X202	UP } LIFTER RIGHT		CYLINDER
	03	X203	DOWN }		SENSOR
	04	X204	LEFT } BELT EXTEND		CYLINDER
	05	X205	RIGHT }		SENSOR
	06	X206			
	07	X207			
	08	X208			
	09	X209			
	10	X210	FRONT } CUTTER	} CENTRE	CYLINDER
	11	X211	BACK }		

100CH	00	Y10000	START SW LIGHT		
(OUTPUT)	01	Y10001	STOP SW LIGHT		
	02	Y10002	MAIN VALVE RELAY		
	03	Y10003			
	04	Y10004			
	05	Y10005			
	06	Y10006			
	07	Y10007			
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
101CH (OUTPUT)	00	Y10100	LIFTER LEFT		VALVE
	01	Y10101	LIFTER RIGHT		VALVE
	02	Y10102	PULLER LEFT		VALVE
	03	Y10103	PULLER RIGHT		VALVE
	04	Y10104	TOP PRESS LEFT		VALVE
	05	Y10105	TOP PRESS RIGHT		VALVE
	06	Y10106	CRIMP LEFT		VALVE
	07	Y10107	CRIMP RIGHT		VALVE
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V
102CH (OUTPUT)	00	Y10200	BELT EXTEND		VALVE
	01	Y10201	CRIMP CENTRE LEFT	} CENTRE	VALVE
	02	Y10202	CRIMP CENTRE RIGHT		VALVE
	03	Y10203	CUTTER		VALVE
	04	Y10204	CUTTER F/B		VALVE
	05	Y10205	CLAMP BELT LEFT		VALVE
	06	Y10206	CLAMP BELT RIGHT		VALVE
	07	Y10207			
	COM	0Vdc	GND		0V
	COM	0Vdc	GND		0V

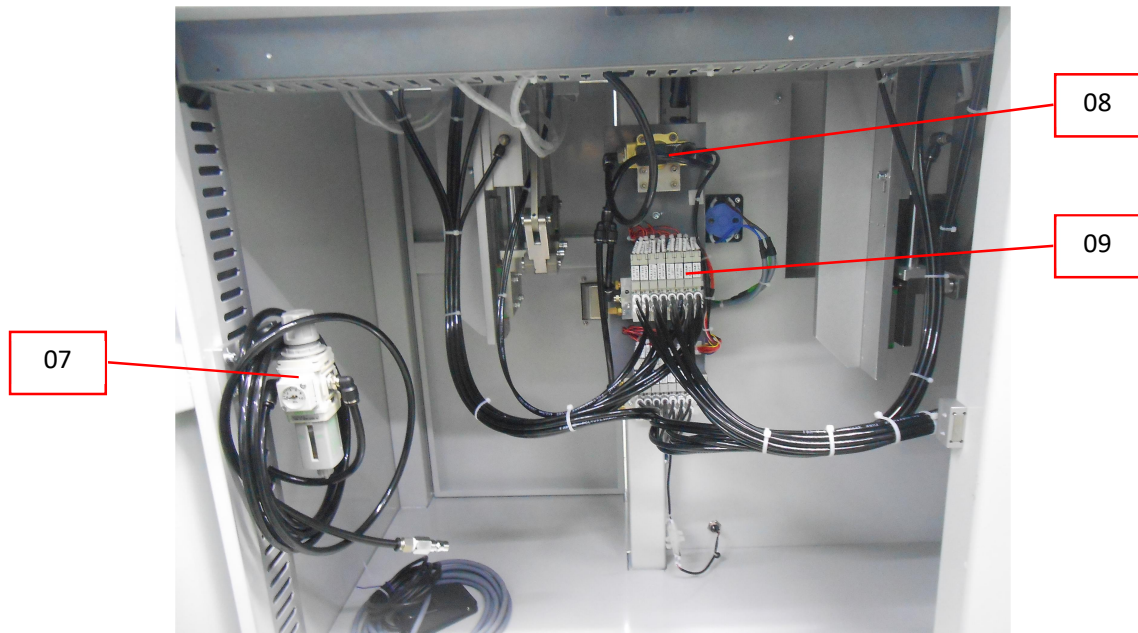
CP1W-8ED

103CH (INPUT)	00	X300	UP	}	CLAMP BELT L	}	CENTRE	CYLINDER SENSOR
	01	X301	DOWN					
	02	X302	UP	}	CLAMP BELT R			CYLINDER SENSOR
	03	X303	DOWN					
	04	X304	UP	}	CRIMP CENTRE L			CYLINDER SENSOR
	05	X305	DOWN					
	06	X306	UP	}	CRIMP CENTRE R			CYLINDER SENSOR
	07	X307	DOWN					
		COM	24Vdc	24V Power supply				

RS232C [D-SUB9P(F)]	(COM 1) TOUCH SCREEN	PLC (RS232 Port)
2	↔	RD
3	↔	SD
4	□	RS
5	□	CS
6		
9	↔	SG



ITEM No.	DESCRIPTION
01	MINIATURE CIRCUIT BREAKER
02	PLC
03	DC POWER SUPPLY
04	RELAY
05	CONNECTOR L
06	CONNECTOR S



ITEM No.	DESCRIPTION
07	AIR REGULATOR
08	MAIN SOLENOID VALVE
09	SOLENOID VALVE WITH MANIFOLD